

Andrew Lee

Software Engineer

Taiwan

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Experience

Wipro Taiwan, supporting Google

New Taipei City, Taiwan

Modeling Engineer

Jun 2026 - Now

- Refactored C++ ISP simulation model blocks through a framework / inheritance swap while keeping I/O and behavior identical; shipped well ahead of deadline with same-day LGTMs.
- Built a Python utility that dumps function inputs and outputs to Protobuf and checks Python-to-C++ ports with hypothesis tests, so refactors and ports match bit-exact on generated cases instead of by eye.
- Built one TUI over several Directed Pattern tools, with structured logs, so a run can be followed end to end.
- Ported urgent Python simulation functions to C++ on a 2-week milestone and matched original behavior.

MOXA Inc.

New Taipei City, Taiwan

IT Application Engineer

Jun 2024 - Nov 2025

- Authored frontend engineering guide, component library with full testing and automated publishing pipeline, standardizing practices and reducing new feature delivery time from multiple weeks to 1 month across the IT department.
- Led frontend development for Partner Zone platform serving global distributors and partners, delivering consistent, high-performance UIs that improved cross-regional user experience and adoption.

Athena Information Systems

Taipei City, Taiwan

Senior Software Engineer

Oct 2023 - Mar 2024

- Led complete Vue 2 to Vue 3 migration, modernizing architecture and improving long-term maintainability and developer velocity.

PhaseTech & Advanced Computation Laboratory

New Taipei City, Taiwan

Technical Lead / Founding Engineer

Jan 2019 - Jul 2023

- Designed and built internal GIS framework on CesiumJS, reducing development time and increasing code reuse across applications.
- Architected and rebuilt DeepSea9v4 global fishery monitoring system, achieving 75% memory reduction and 90% render performance improvement over previous generation.
- Led end-to-end development of MarineCom ([Patented TW202437745A](#)), delivering an affordable satellite communication solution for maritime users.

Projects

Open Bodycam

<https://open-bodycam.xtl.tw>

Creator & Maintainer

2026 - Present

- Designed, prototyped, and open-sourced a low-cost wearable body camera system with custom 3D-printed enclosure for transparent, hackable real-world use.
- Optimized CMA memory allocation + real-time vectorized NumPy alpha blending, achieving stable 1080p overlay compositing on severely constrained embedded hardware (512MB RAM, limited CPU, no GPU).
- Delivered full-stack solution spanning embedded Linux (Picamera2, libcamera, OpenCV), backend, and React/Vite frontend.

Shutter Sim

<https://shutter-sim.xtl.tw>

Creator & Maintainer

2026 - Present

- Built an educational reverse ISP that walks a JPEG back through inverse gamma, CCM, GBGR remosaic, white balance, shading, defects, and black level to a synthetic Bayer plane. Not a RAW recovery.

Customized Aerial Screen Saver

<https://github.com/FalconLee1011/Customized-Aerial-Screen-Saver>

Creator

2024 - 2025

- Reverse-engineered protected system assets and developed macOS Swift/SwiftUI app for deep customization of Apple's Aerial screen saver, reaching 230+ GitHub stars and featured in [MacRumors community](#).

Skills

Languages	Python, C++, TypeScript/JavaScript, Swift
Systems & Simulation	Python tooling, Protobuf, bit-exact validation, Linux
Embedded Systems	Raspberry Pi, libcamera / Picamera2, OpenCV, NumPy, 3D CAD / printing
Infrastructure	Git, Cloudflare, Docker, GitLab CI, Flask, RESTful APIs
Frontend	Vue, Nuxt, Vite, React